

MATH 195: 9/24

RADICAL FUNCTIONS

A *radical function* is an inverse of a power function. We write $\sqrt[n]{x}$ for the inverse of x^n .

One use of radicals is to solve equations involving power functions. Namely, if you have an equation in the form $A^n = B$ then you hit both sides with a radical:

$$\sqrt[n]{A^n} = \sqrt[n]{B}$$

$$A = \sqrt[n]{B}.$$

This is used, for instance, to find x -intercepts.

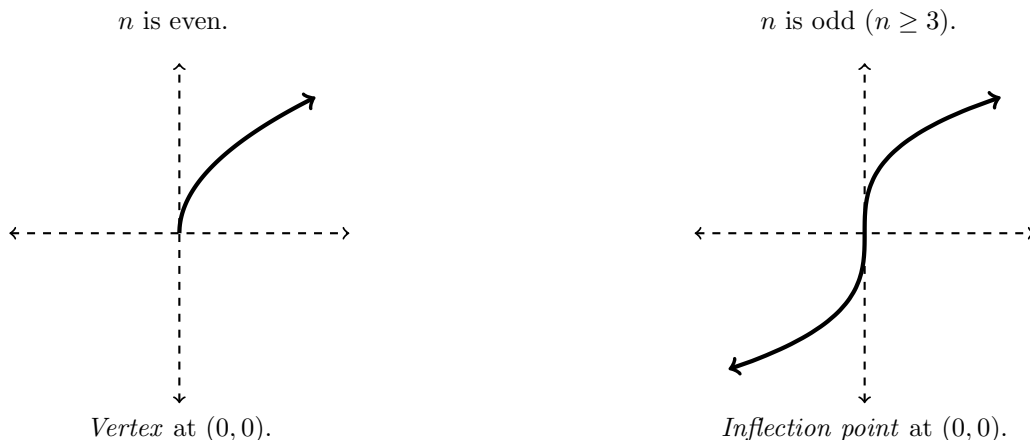
- (1) Solve the equation $(x + 4)^3 = 8$.
- (2) Find the x -intercept of $f(x) = (x - 2)^3 + 1$.
- (3) Find the x -intercepts of $g(x) = (x - 1)^4 - 16$.

For the last problem you should've found two x -intercepts, which illustrates an issue. Some power functions aren't one-to-one, and so don't have inverses!

- If n is odd, then x^n is increasing everywhere, so it is one-to-one, so it has an inverse. There's no problem in this case.
- If n is even, then x^n is not one-to-one. So it doesn't have an inverse. Imagine reflecting the graph across the diagonal line $y = x$; the resulting curve fails the vertical line test.
- To get an inverse, we have to *restrict the domain*. We limit the inputs of the function to a smaller domain on which it is one-to-one. For x^n with even n , the traditional choice is to restrict the domain to $[0, \infty)$.
- Phrased in more direct language, $\sqrt[n]{x}$ for even n is defined to be the positive root.

This complication will come back. When we study trig functions, they will not be one-to-one and we will have to restrict the domain to get inverses.

We get graphs for $\sqrt[n]{x}$ by reflecting the graphs of x^n across the diagonal line $y = x$. The shape depends on the parity of n .



When sketching a graph of a geometric transformation of a power function, there are two more important details:

- (1) Where is the vertex/inflection point? This is answered by identifying the horizontal and vertical translations.
- (2) What is the orientation? For odd n , the function $\sqrt[n]{x}$ is an odd function (= the graph has rotational symmetry), so we only have worry about two orientations. For even n , there are four orientations: no reflection, only a vertical reflection, only a horizontal reflection, and both reflections.

Because we now have to worry about horizontal transformations besides translations, a new complication arises. With horizontal translations, everything happens backward. This also means we want to do multiplication versus addition in the backward order to read off the transformations. For example, if the input of a function is $2x + 4$ we want to rewrite it as $2(x + 2)$, with the addition happening first, to read off a horizontal translation by 2.

- (1) Sketch a graph of $a(x) = \sqrt{2 - x} + 1$. Explain why the vertex is at the point $(2, 1)$. Based on the graph, what are the domain and range of a ?
- (2) Sketch a graph of $b(x) = -2\sqrt[3]{4x + 4}$. What are its domain and range? Where is it increasing? Decreasing? Concave up? Concave down?
- (3) Sketch graphs of all four orientations for reflections of $c(x) = \sqrt[4]{x}$. Write formulas for all four of them.
- (4) Sketch a graph of $d(x) = 3\sqrt[4]{x - 1} - 3$. Find its intercepts.
- (5) Find the intercepts of $f(x) = -(2x + 3)^3 - 4$.
- (6) Find the intercepts of $g(x) = \frac{2}{(x - 1)^2} - 4$.
- (7) Determine the domain of $h(x) = \sqrt{3x - 4}$ by solving the inequality $3x - 4 \geq 0$. Why does solving this inequality tell you the domain of h ?
- (8) Determine the domain of $j(x) = 4\sqrt[5]{2 - 5x} - 1$ without graphing.